Model Specification

POLS 602
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Review

1. What is the difference between an estimand and an estimator?
2. What does a p-value tell us, and what does it not tell us?
3. How does an F-test compare a restricted and an unrestricted model?

Today

- What does it mean for a parameter to be identified?
- How do we adapt a linear model to different data generating processes?
- How do we model groups, time, treatments, thresholds, and changing coefficients?

Identification

What Is Identification?

- Estimation asks how to compute a number from data
- Inference asks how uncertain that number is
- **Identification**: whether the parameter can be recovered from the data at all

A parameter is identified if only one value of it is consistent with the data, even if we observed the entire population.

Counting Equations

- $a + b = 4$: one equation, two unknowns, many solutions
- $a + b = 4$ and $a - b = 2$: two equations, two unknowns, one solution ($a = 3, b = 1$)
- $a + b = 4$ and $2a + 2b = 8$: two equations, but the second repeats the first, so many solutions
- $a + b = 4$, $a - b = 2$, and $a = 5$: more equations than unknowns, and no solution satisfies all three

An Identification Condition

An **identification condition** is an assumption about the population that gives us equations the true parameters must satisfy.

For OLS, the assumption is that the true error is uncorrelated with every regressor:

$$\mathbb{E}\left[\mathbf{x}_i \, \varepsilon_i\right] = \mathbf{0}$$

$\mathbf{x}_i$ has k entries, so this is k equations, one per regressor:

$$\mathbb{E}[x_{1i}\varepsilon_i] = 0, \quad \mathbb{E}[x_{2i}\varepsilon_i] = 0, \quad \ldots, \quad \mathbb{E}[x_{ki}\varepsilon_i] = 0$$

Two Coefficients, Two Equations

Simple regression: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

The regressors are $\mathbf{x}_i = (1,\ x_i)$, so the identification condition $\mathbb{E}[\mathbf{x}_i \varepsilon_i] = \mathbf{0}$ gives two equations:

$$\text{Constant: } \mathbb{E}[\varepsilon_i] = 0 \qquad\qquad x\text{: } \mathbb{E}[x_i \varepsilon_i] = 0$$

Substitute $\varepsilon_i = y_i - \beta_0 - \beta_1 x_i$:

$$\mathbb{E}[y_i - \beta_0 - \beta_1 x_i] = 0 \qquad\qquad \mathbb{E}[x_i (y_i - \beta_0 - \beta_1 x_i)] = 0$$

Solving for the Intercept

Take the constant equation and expand it:

$$\mathbb{E}[y_i] - \beta_0 - \beta_1 \, \mathbb{E}[x_i] = 0$$

Solve for β_0:

$$\beta_0 = \mathbb{E}[y_i] - \beta_1 \, \mathbb{E}[x_i]$$

Solving for the Slope

Put that β_0 into the x equation:

$$\mathbb{E}[x_i y_i] - (\mathbb{E}[y_i] - \beta_1 \mathbb{E}[x_i])\, \mathbb{E}[x_i] - \beta_1\, \mathbb{E}[x_i^2] = 0$$

Group the terms with and without β_1:

$$(\mathbb{E}[x_i y_i] - \mathbb{E}[x_i]\, \mathbb{E}[y_i]) - \beta_1 \left(\mathbb{E}[x_i^2] - \mathbb{E}[x_i]^2\right) = 0$$

The first group is $\mathrm{Cov}(x_i, y_i)$ and the second is $\mathrm{Var}(x_i)$, so

$$\beta_1 = \frac{\mathrm{Cov}(x_i, y_i)}{\mathrm{Var}(x_i)}$$

From Population to Sample

- The equations use population expectations, which we never observe
- Replace each expectation with its sample average

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0$$

$$\frac{1}{n}\sum_{i=1}^{n} x_i\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0$$

The Normal Equations Again

All k equations at once:

$$\frac{1}{n}\mathbf{X}^\top\left(\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\right) = \mathbf{0}$$

Multiply by n and distribute: $\mathbf{X}^\top\mathbf{X}\,\widehat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{Y}$

Exactly Identified

- Two unknowns and two equations, so one solution
- The solution exists as long as $\mathrm{Var}(x_i) \neq 0$
- If x_i is related to the error, we get the same formulas, but they are not the true coefficients

Under-Identified

- **Under-identified**: fewer independent equations than unknown coefficients
- Many coefficient values satisfy the equations, so the data cannot pick one
- More data does not help
- In regression, perfect multicollinearity causes this

Perfect Multicollinearity

Suppose $x_{2i} = 2x_{1i}$ in the model $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$.

$$\mathbb{E}[x_{2i}\varepsilon_i] = \mathbb{E}[2x_{1i}\varepsilon_i] = 2\,\mathbb{E}[x_{1i}\varepsilon_i]$$

- The equation for x_2 is the equation for x_1 multiplied by 2
- Two unknowns but only one independent equation

The model is $y_i = (\beta_1 + 2\beta_2)\, x_{1i} + \varepsilon_i$, so only $\beta_1 + 2\beta_2$ is identified.

Over-Identified

- More equations than unknowns
- Each subset of equations would give its own answer
- Counting example: $a + b = 4$ and $a - b = 2$ give $a = 3$, but $a = 5$ says otherwise
- If the answers disagree, at least one assumption is wrong

From Equations to Causal Effects

- The equations only hold if the identification condition holds
- If something omitted affects Y and is related to X, the error is correlated with X and the condition fails
- The coefficient is then not the effect we want, however much data we have
- **Causal identification**: arguing that the condition holds, so that the coefficient equals the causal effect

The Plan for Today

Each specification comes with an assumption that makes the identification condition plausible.

Specification	Identifying assumption
Fixed effects	Confounders do not change over time
Difference-in-differences	Parallel trends
Discontinuities and kinks	Nothing else jumps at the cutoff
Structural breaks	Coefficients are stable across periods

Fixed Effects

Panel Data

- **Panel data**: the same units observed in several time periods
- Each row is one unit in one period, so each unit appears in several rows
- y_{it} is the outcome for unit i in period t

Country (i)	**Year (t)**	**Growth (y_{it})**	**Democracy (x_{it})**
A	2000	2.1	6
A	2001	2.4	7
B	2000	4.0	3
B	2001	3.7	3

Unobserved Heterogeneity

- Units differ in ways we cannot measure
- Some of those differences are fixed over time
- If they relate to X, they bias $\hat{\beta}$

How can we control for a variable we never observe?

The Fixed Effects Model

$$y_{it} = \beta x_{it} + \alpha_i + \varepsilon_{it}$$

- i indexes units and t indexes time periods
- α_i is a separate intercept for each unit, containing everything about the unit that is fixed over time

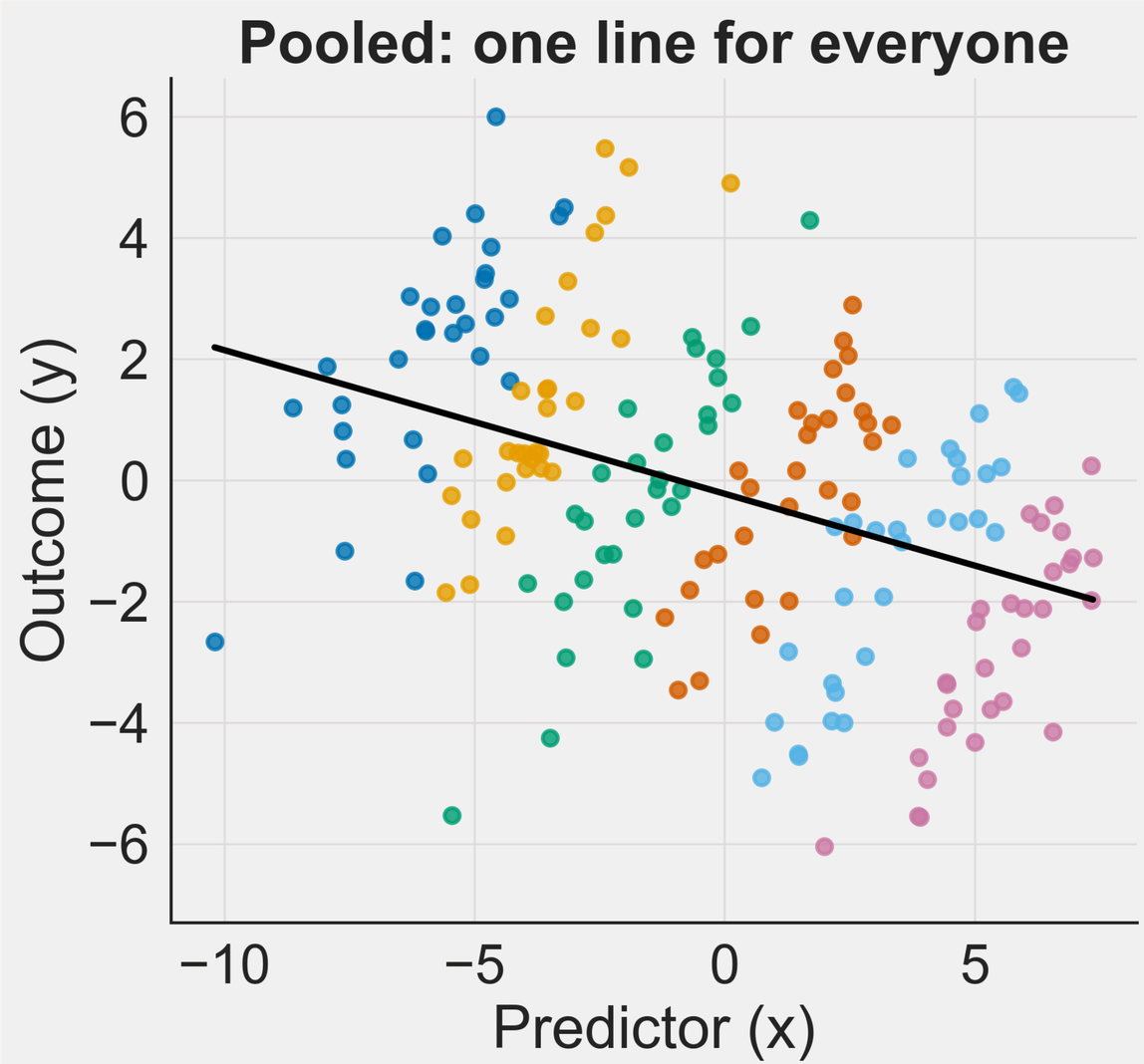
Compare to the ordinary model, where one intercept β_0 is shared by everyone.

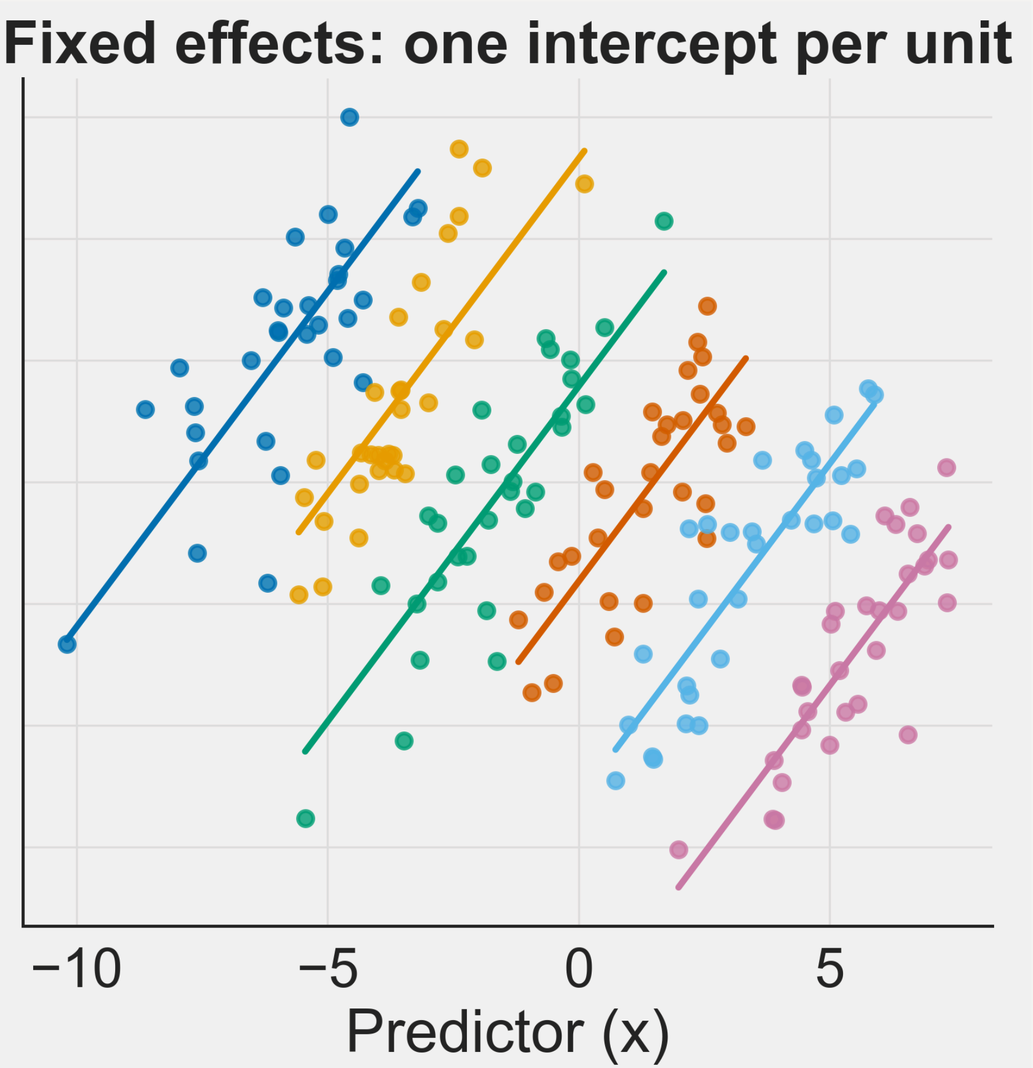
Dummy Variable Implementation

$$y_{it} = \beta x_{it} + \sum_{j=1}^{N} \alpha_j D_{j,it} + \varepsilon_{it}$$

- $D_{j,it} = 1$ if observation it belongs to unit j
- One dummy per unit, and no common intercept

Pooled OLS Can Get the Sign Wrong





Fixed Effects Summary

- Fixed effects compares each unit to itself over time
- Only variation within a unit identifies β
- Variables that never change within a unit cannot be estimated
- Identification condition: x_{it} is uncorrelated with ε_{it}, after α_i is accounted for
- Confounders that are constant within a unit are removed, measured or not
- Confounders that change over time are still in ε_{it}

Categorical Variables

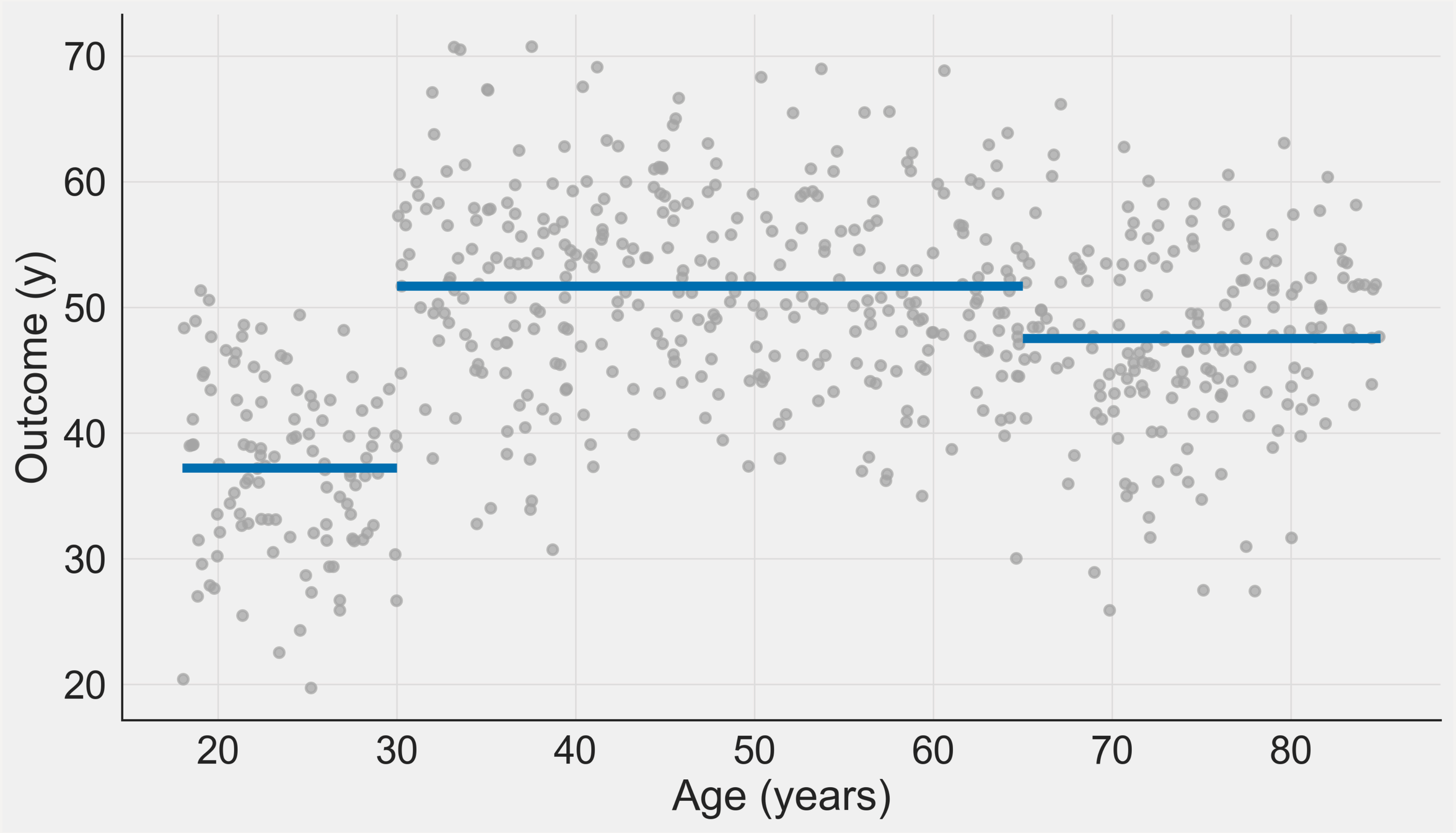
Thresholds as Categories

- A cutoff, such as a legal age, suggests a step in the outcome
- Replace a continuous variable with category dummies
- Leave one category out as the baseline
- Each coefficient is a difference from the baseline

$$y_i = \beta_0 + \beta_1 D_{30\text{-}64,i} + \beta_2 D_{65+,i} + \varepsilon_i$$

- Here the baseline is under 30, so β_0 is its mean
- β_1 and β_2 are differences from that mean

Group Means as a Step Function



Time Period Dummies

$$y_{it} = \beta x_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

- γ_t is a separate intercept for each time period
- Absorbs anything that hits every unit equally in a period
- Notice α_i, a unit fixed effects intercept. Paired with γ_t, this is a two-way fixed effects model
- One unit dummy and one time dummy must be dropped, as with any set of dummies plus an intercept

Difference-in-Differences

Snow and the Cholera Outbreaks

- Cholera killed thousands of Londoners in 1849 and again in 1854
- Most doctors believed it spread through bad air
- John Snow argued that it spread through contaminated water
- *How could Snow test this without running an experiment?*

A Natural Experiment

- Two companies supplied London homes with Thames water
- In 1852, Lambeth moved its water intake upstream, above London’s sewage
- Southwark & Vauxhall did not move its intake
- Customers of the two companies lived on the same streets

Two Bad Comparisons

- Lambeth before vs. after 1852:
  - No between group comparison. Infection rates could be the same across London.
- Lambeth vs. Southwark & Vauxhall in 1854:
  - No time comparison that isolates the change from Lambeth. Maybe the people who get water from each supplier differ in other ways.
- Each comparison alone mixes the effect of the water with something else

Four Means

	1849 (before)	1854 (after)
Southwark & Vauxhall	$\bar{y}_{S,0}$	$\bar{y}_{S,1}$
Lambeth	$\bar{y}_{L,0}$	$\bar{y}_{L,1}$

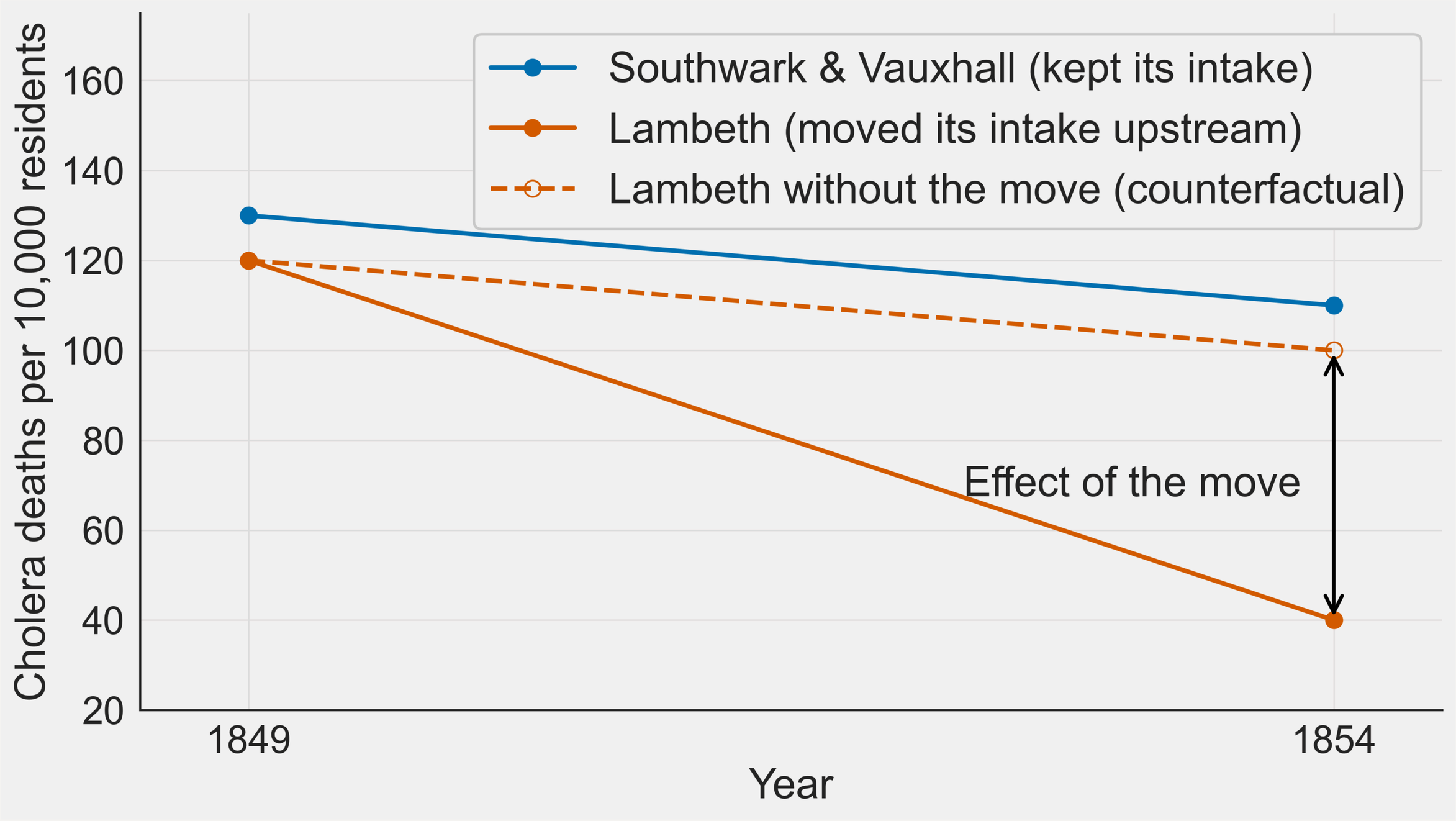
$$\widehat{\delta} = (\bar{y}_{L,1} - \bar{y}_{L,0}) - (\bar{y}_{S,1} - \bar{y}_{S,0})$$

What is the identifying assumption here?

Identifying Assumptions

- Parallel Trends: Without the treatment, the treated group and the control group would have followed similar trajectories over time.
- SUTVA (Stable Unit Treatment Value Tax/Assumption): The treatment applied to one unit does not spill over or affect the outcome of other units
- No Anticipation Effects: The treatment group does not change its behavior before the intervention actually occurs because they expect it in the future

Parallel Trends



The Regression

$$y_i = \beta_0 + \beta_1 \text{Lambeth}_i + \beta_2 \text{Post}_i + \beta_3 (\text{Lambeth}_i \times \text{Post}_i) + \varepsilon_i$$

- $\text{Lambeth}_i = 1$ if the neighborhood is supplied by Lambeth
- $\text{Post}_i = 1$ if the observation is from 1854
- β_1: the gap between the groups in 1849
- β_2: the change over time for Southwark & Vauxhall
- β_3: the difference-in-differences estimate

Why β_3 Is the Estimate

	1849	1854	Change
Southwark & Vauxhall	β_0	$\beta_0 + \beta_2$	β_2
Lambeth	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_2 + \beta_3$

Lambeth = 1 only for Lambeth, and Post = 1 only in 1854.

Difference in the changes: $(\beta_2 + \beta_3) - \beta_2 = \beta_3$

Simulated Cholera Example

- Southwark & Vauxhall: 129.5 deaths per 10,000 in 1849, 109.5 in 1854
- Lambeth: 119.8 deaths per 10,000 in 1849, 39.7 in 1854
- $\hat{\beta}_3 = (39.7 - 119.8) - (109.5 - 129.5) = -60.1$

The true effect in the simulation is -60.

DiD as Fixed Effects

$$y_{it} = \delta D_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

- $D_{it} = 1$ if unit i is treated in period t
- In the cholera example, $D_{it} = \text{Lambeth}_i \times \text{Post}_t$
- Unit and time fixed effects replace the group and period dummies
- Works with many units and many periods

Kinks and Discontinuities

Treatment by a Cutoff

- Treatment is assigned by whether a score, the running variable, crosses a threshold
- Examples: a vote share above 50%, a test score above a passing grade, an age past eligibility
- *Compare units just above and just below the cutoff*

Regression Discontinuity

$$y_i = \beta_0 + \beta_1(x_i - c) + \beta_2 D_i + \beta_3 D_i(x_i - c) + \varepsilon_i$$

- x_i is the running variable, c is the cutoff, and $D_i = 1$ if $x_i \geq c$
- β_2 is the jump in the outcome at the cutoff
- β_3 lets the slope differ on each side

Two Lines, One Regression

Below the cutoff ($D_i = 0$):

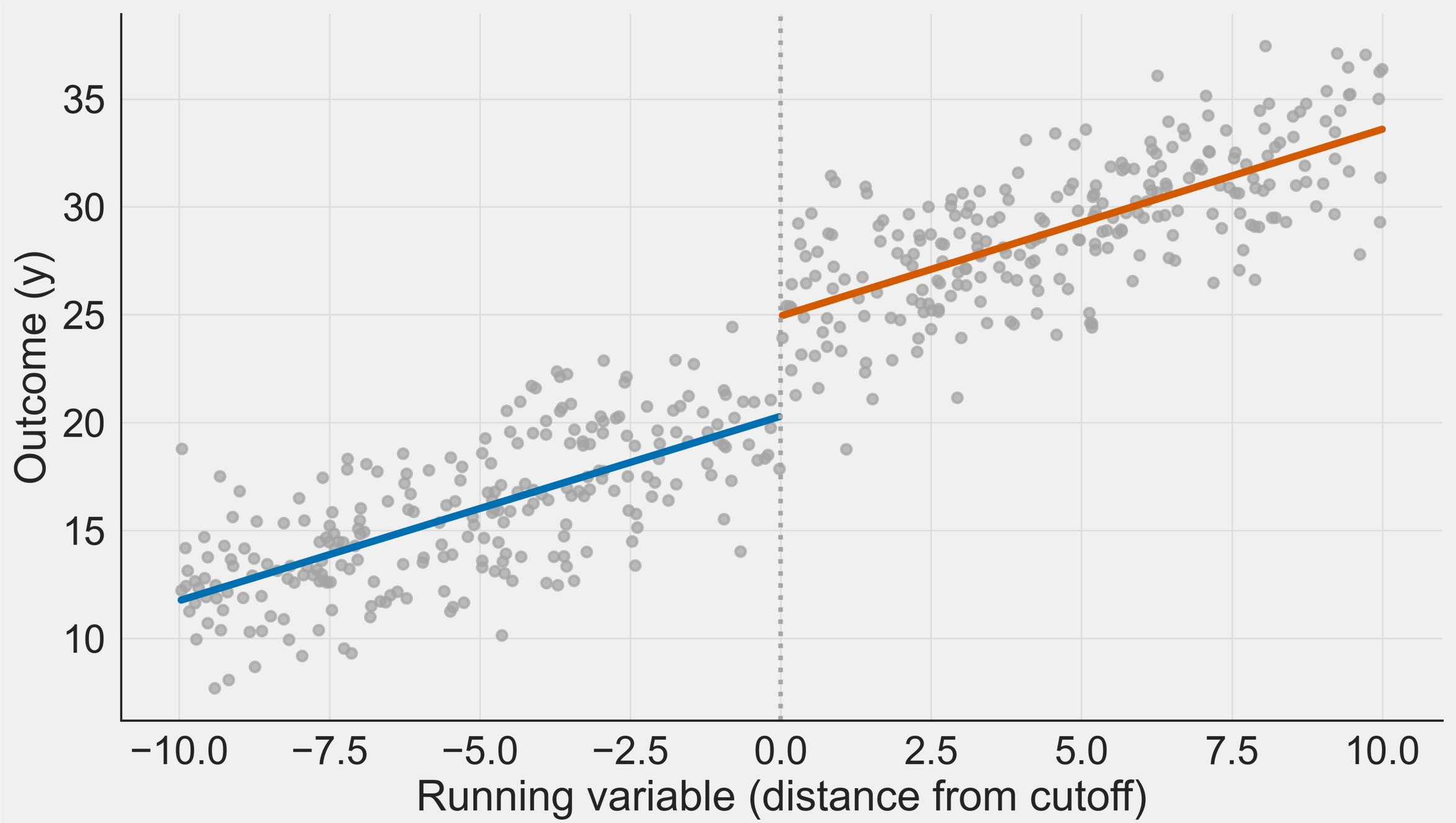
$$y_i = \beta_0 + \beta_1 (x_i - c) + \varepsilon_i$$

At or above the cutoff ($D_i = 1$):

$$y_i = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)(x_i - c) + \varepsilon_i$$

At $x_i = c$ the two lines differ by β_2.

A Jump at the Cutoff



The Identifying Assumption

- Units cannot precisely control their score
- Nothing else changes sharply at the cutoff
- Only the treatment jumps
- Units just above and just below the cutoff are comparable, as if treatment were randomized

Bandwidth

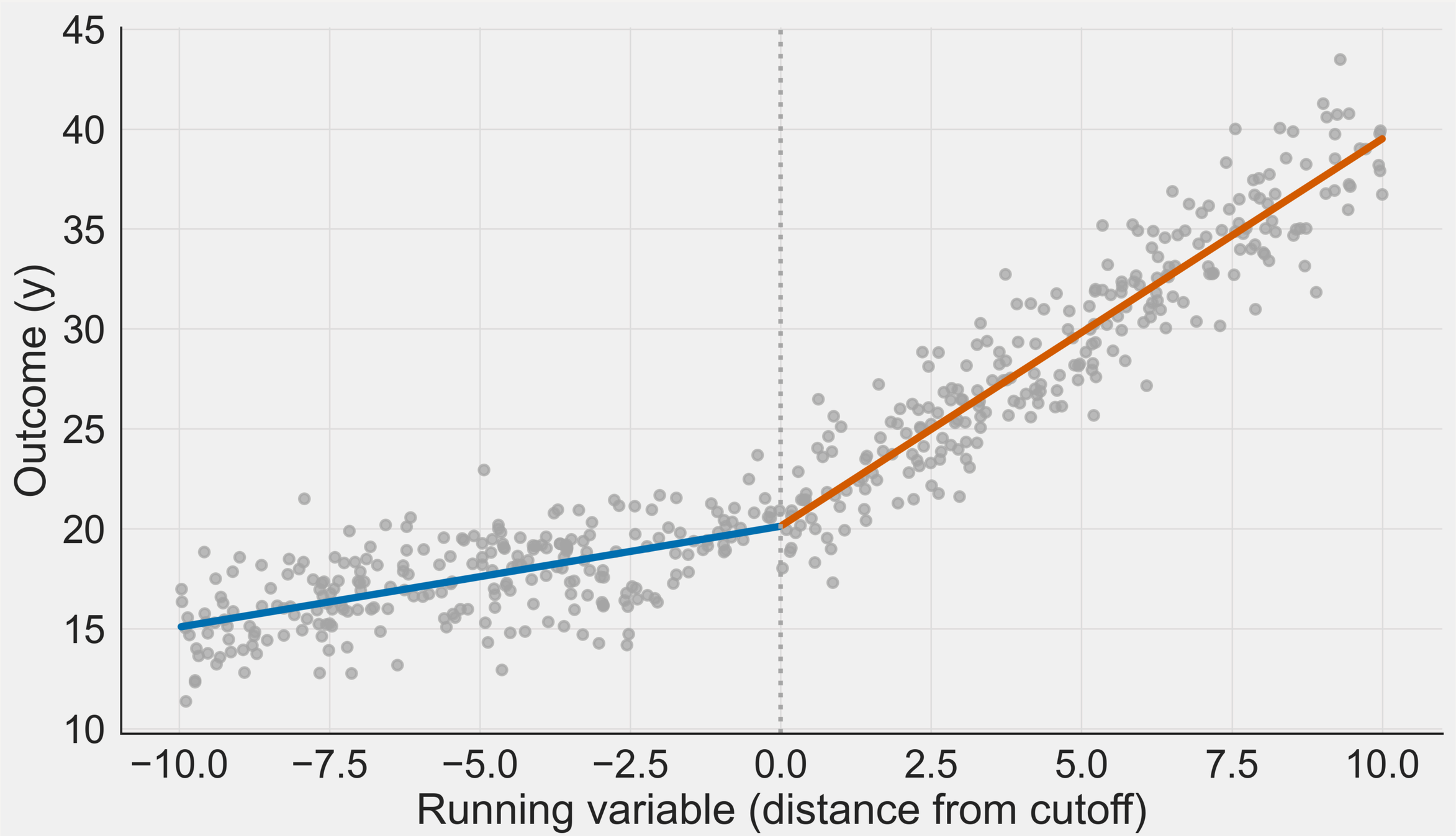
- **Bandwidth**: the window around the cutoff used for estimation
- Narrow window: more comparable units, fewer observations
- Wide window: more observations, but the straight lines must fit over a longer range
- Report results across several bandwidths

Regression Kink

$$y_i = \beta_0 + \beta_1(x_i - c) + \beta_2 D_i(x_i - c) + \varepsilon_i$$

- $D_i = 1$ if $x_i \geq c$, as before
- Below the cutoff the slope is β_1, above it is $\beta_1 + \beta_2$
- No jump in the level
- β_2 is the change in slope at the cutoff

A Change in Slope



Structural Breaks

Do the Parameters Change?

- A policy change, a war, or a new technology can change how X relates to Y
- A structural break is a change in the regression coefficients
- *Is one set of coefficients enough for the whole sample?*

Restricted and Unrestricted Models

Restricted: the same coefficients in both periods

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

Unrestricted: different coefficients in each period

$$y_t = \begin{cases} \beta_0^{(1)} + \beta_1^{(1)} x_t + \varepsilon_t & t < t^* \\ \beta_0^{(2)} + \beta_1^{(2)} x_t + \varepsilon_t & t \geq t^* \end{cases}$$

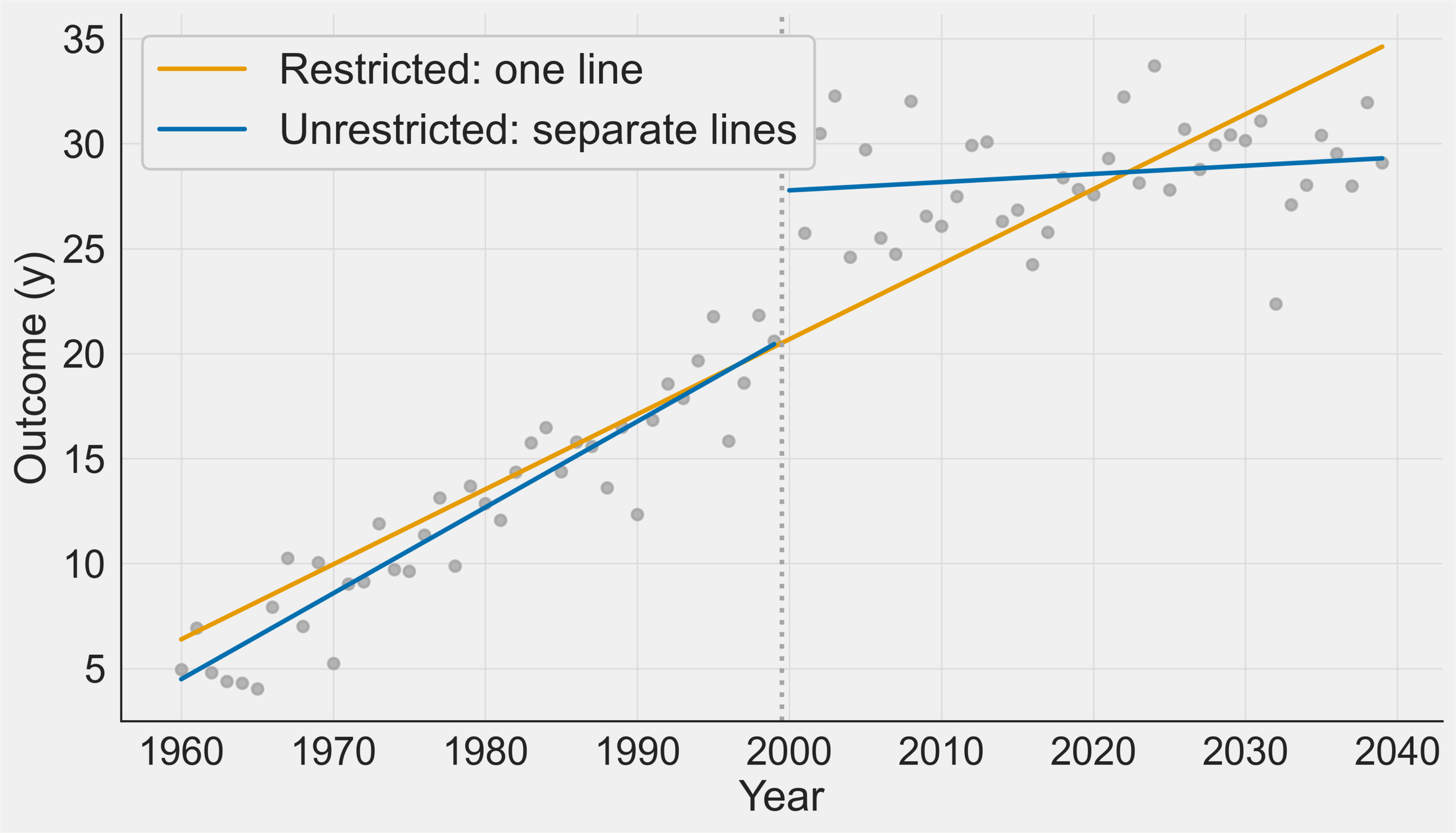
The Chow Test

$$F = \frac{(\text{SSE}_R - (\text{SSE}_1 + \text{SSE}_2))/k}{(\text{SSE}_1 + \text{SSE}_2)/(n - 2k)}$$

- SSE_R : the sum of squared errors from the pooled regression
- $\text{SSE}_1, \text{SSE}_2$: the sums of squared errors from the separate regressions
- Under the null, $F \sim F_{k,\, n-2k}$

Same formula as lecture 7, with $\text{SSE}_U = \text{SSE}_1 + \text{SSE}_2$, $q = k$, and $2k$ coefficients in the unrestricted model.

A Break in the Relationship



The Chow Test as an Interaction

Let $D_t = 1$ if $t \geq t^*$. Interact every regressor with D_t:

$$y_t = \beta_0 + \beta_1 x_t + \gamma_0 D_t + \gamma_1 D_t x_t + \varepsilon_t$$

- Before the break: intercept β_0, slope β_1
- After the break: intercept $\beta_0 + \gamma_0$, slope $\beta_1 + \gamma_1$
- $H_0 : \gamma_0 = \gamma_1 = 0$, tested with an F-test of two restrictions

Choosing the Break Date

- The Chow test assumes we know when the break happened
- It does not work if we pick the date after looking at the data
- With an unknown date, test every candidate date and take the largest F

Summary

- A parameter is identified if the data can pin down a unique value, and every specification rests on an identifying assumption
- Dummy variables let units, time periods, and ranges of a variable have their own intercepts, and fixed effects remove unit-level confounders that do not change over time
- Difference-in-differences and discontinuity designs identify causal effects from an interaction and a jump or kink, under parallel trends and continuity
- A structural break test asks whether all the coefficients differ across periods, and the Chow test is an F-test of a restricted against an unrestricted model

Key Definitions

Key Definitions

- **Identification**: whether a parameter can be recovered uniquely from the distribution of the data
- **Identification condition**: an assumption about the population that gives equations the true parameters must satisfy
- **Exactly identified**: as many independent equations as parameters
- **Under-identified / over-identified**: fewer independent equations than parameters / more equations than parameters
- **Causal identification**: the argument that the identification condition holds, so the coefficient is the causal effect

Key Definitions

- **Panel data**: the same units observed in several time periods
- **Unobserved heterogeneity**: differences between units that we cannot measure
- **Fixed effects**: a separate intercept for each unit, absorbing everything about it that is constant over time
- **Pooled OLS**: one regression on all rows, ignoring which unit each row comes from
- **Within estimator**: OLS on data with each unit's mean subtracted

Key Definitions

- **Reference category**: the omitted group that dummy coefficients are measured against
- **Two-way fixed effects**: unit and time period intercepts together
- **Difference-in-differences**: the change in treated units minus the change in control units
- **Parallel trends**: without treatment, treated and control groups would have changed by the same amount
- **Running variable**: the score that determines treatment assignment

Key Definitions

- **Regression discontinuity**: a jump in the outcome at a cutoff in the running variable
- **Regression kink**: a change in the slope of the outcome at a cutoff
- **Bandwidth**: the window around the cutoff used for estimation

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Key Definitions

- **Structural break**: a change in the regression coefficients at some point in the sample
- **Chow test**: an F-test comparing a pooled model to separate models for each period
- **Sup-F test**: a break test that searches over all candidate break dates